Environmental Control System (Mod 6/2014) – General Data

1. General

The temperature control system provides manual regulation of temperature inside the aircraft, by controlling the flow of engine compressor bleed air entering the heating system.

2. Description

Components of the temperature control system are: a manual control panel, outside air temperature sensor, duct temperature sensor, duct overheat switch and duct overheat caution message displayed through CAS.

A. Control Panel

The temperature control system control panel is located on the instrument and identified ENVIRONMENTAL. For heating there are two switches – a MODE switch with an upper ON and lower OFF position; and a TEMP switch which is a 3 position switch, spring-loaded to the center neutral position, with momentary WARM and COOL positions at the upper and lower points.

If the MODE switch is placed in the OFF position, the bleed air valve that supplies bleed air to the ejector assembly will motor to the closed position. If the MODE switch is placed in the ON position, the pilot can then adjust the position of the valve as desired by holding the TEMP switch in either the up (WARM) or down (COOL) position.

B. Outside Air Temperature Sensor

The outside air temperature sensor is a probe located in the cooling system left hand side of the aircraft near the forward baggage compartment door at station 347.00 to 376.00.

C. Cabin Temperature Sensor

The cabin temperature sensor is located in a plastic housing mounted on the mid cabin, approximately at station 271.90, in a multi purpose bracket that houses one of the cabin dome lights and PA speakers.

D. Duct Overheat Switch

The duct overheat switch is a probe located in the heating system silencer.

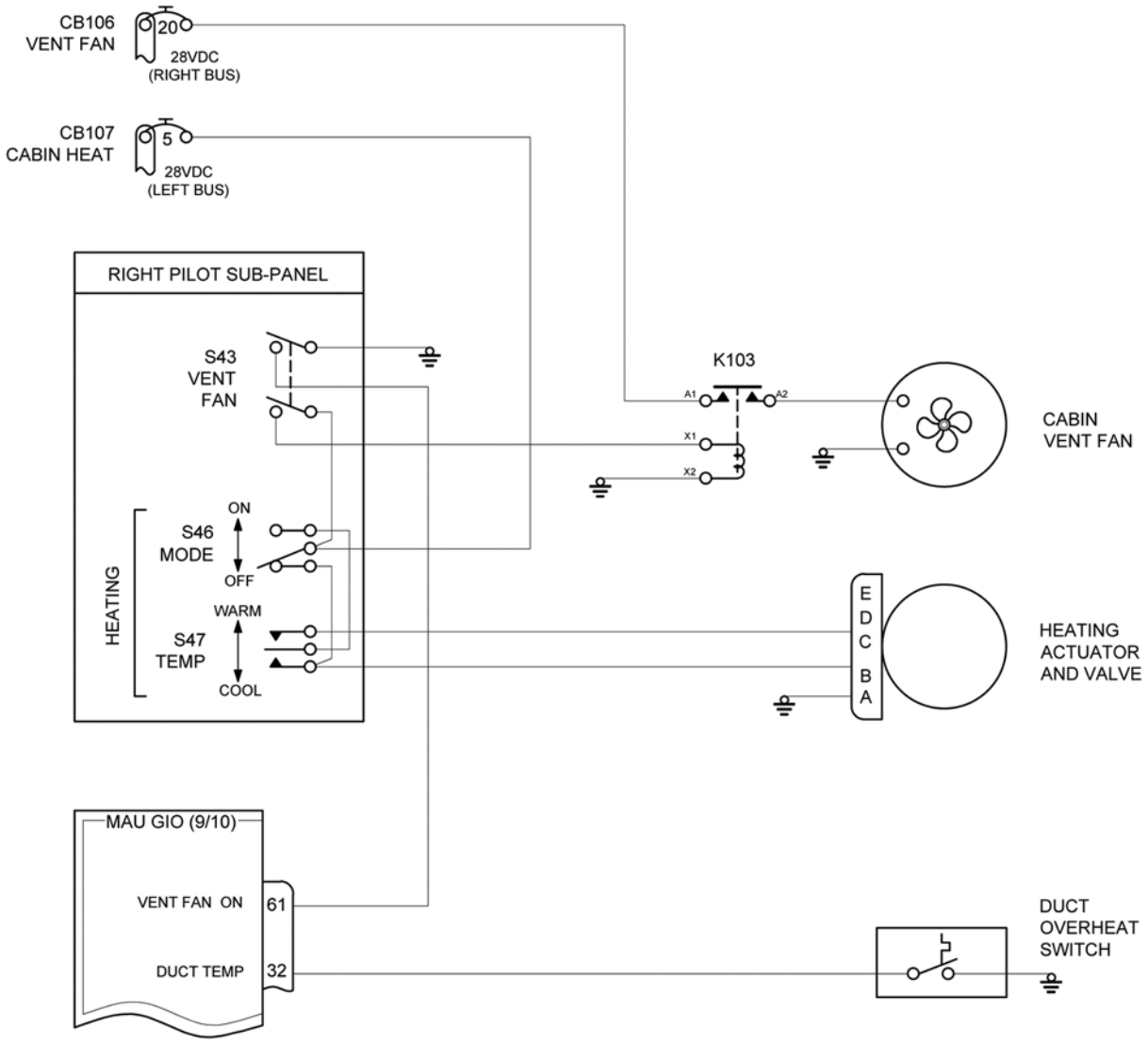
E. Duct Overheat Caution Message

The duct overheat caution message is displayed on the CAS message window and is identified Duct Overheat.

3. Operation

Refer to Figure 1 for details.

The hot air valve is driven to the desired position by the selection of the HEATING–MODE and HEATING–TEMP switches, which routes 28 Volt DC direct to the appropriate hot air valve actuator windings, in the COOL–WARM positions.



Temperature Control Systems – Electrical Schematic
Figure 1

4. Adjustment/Test

A. Test Environmental Control System Controls

Equipment required:

- Source of external electrical power.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) On the instrument panel, select the HEATING–MODE switch to ON and HEATING–TEMP switch to WARM.

3) Check audibly that the hot air valve actuator, located under the floor below the pilot's seat, runs for 30 ± 8 seconds.

4) Select the HEATING–TEMP switch to COOL, and check that hot air valve actuator again runs for 30 ± 8 seconds.

5) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Function Test Temperature Control System

1) With engines running (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data) set one engine to at least 0 percent power (if test is carried out in flight, set engines for level flight).

2) Set LEFT and RIGHT– BLEED AIR switches ON, and check that the cabin air control handle, located in the cockpit center pedestal, is fully pushed in (valve open position).

3) Confirm that temperature output varies appropriately when the HEATING–TEMP switch is selected WARM/COOL.

4) Select HEATING–TEMP switch and BLEED AIR switches to OFF. Stop engines (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

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